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EDUCATION

University of California, Los Angeles (UCLA)

Expected June 2027

Master of Engineering in AI (MEng AI)

Los Angeles, CA

- Incoming Graduate Student; specialized in Artificial Intelligence and Machine Learning.

Purdue University

Dec 2025

B.S. in Computer Science (Software Engineering); Minor in CIT

West Lafayette, IN

- **Selected Coursework:** Data Structures & Algorithms (C++), Operating Systems, Computer Architecture, Software Engineering, Software Testing, Object-Oriented Programming (Java), Programming in C.

EXPERIENCE

Juneng Robotics Co., Ltd.

May 2023 – Aug 2023

Software Engineer Intern

- Built a **React/TypeScript** ops console with tablet-first **PWA** views for on-device perception; added touch-friendly controls, **WebSocket/Apollo GraphQL** streams, and **OpenCV** overlays; cut TTI by 30% and improved operator accuracy by 8 pts.
- Designed a calibration wizard for camera-**LiDAR** fusion; built point-cloud viewers with pinch-zoom/rotate and ROI drawing (**PCL**), with operator QA reviews; cut duplicates/mislabels by 25% and lifted recall by 20% on indoor nav scenes.
- Established a labeling→training workflow with **CVAT** and rater UIs; added augmentation (mosaic/cutout), stratified splits, and HPO; logged runs/artifacts in **Weights & Biases**, wrote model cards, and versioned reproducible configs for QA handoffs.
- Deployed perception services as **ROS** nodes on **C++/TFLite** with a **React** ops **PWA**; added health probes, feature flags, shadow A/B, offline caches and QR pairing, enabling staged rollouts that steadied control loops and improved field UX.

Xiaomi Technology

Jun 2022 – Sep 2022

Software Engineer Intern

- Built and tuned **CTR/CVR** models with **TensorFlow** and **Scikit-learn** (wide&deep, calibrated logistic, **XGBoost**); added a React web UI for cohorts and calibration/lift plots; engineered user/content features improving offline AUC by 4–6%.
- Authored multi-TB **Apache Spark** ETL and exposed API for experiment UI; added aggregations, partitioning, and UDF pruning; ran jobs in **Airflow** and cut payloads/shuffle to lower end-to-end latency by 20% and steady peak traffic.
- Productionized models using **Docker/Kubernetes** and **TensorFlow Serving**; added a React console; integrated **Kafka** events and **Redis** feature cache; shipped drift/latency dashboards (PSI, KS, p50/p95) and **Jenkins** CI with canary flags.

PROJECTS

OfferWise AI – Agentic Offer Decision Platform | *Python, FastAPI, React, LLM Agents, Monte Carlo Simulation*

- Architected an agentic offer-decision platform that extracts compensation, city preferences, and work constraints; built multi-step **LLM** workflows to compare offers using **LifeScore**, **AffordabilityScore**, **FitScore**, and **ConfidenceScore**.
- Engineered a deterministic scoring engine with **feasibility gates**, diminishing income utility, geometric aggregation, and proxy-aware data validation; separated **LLM** orchestration from auditable scoring to reduce hallucination risk.
- Implemented **Monte Carlo** simulations across rent, bonus realization, spending assumptions, and preference weights; generated explainable reports with top drivers, confidence warnings, and offer reversal conditions for decision support.

CommuneKit - Community Engagement Platform | *React, Spring Boot, MySQL, REST APIs, Redis*

- Built a responsive full-stack app with **React** UI and **Spring Boot** services; added server-side pagination, debounced **multi-facet** filters, **URL-state** sync, and optimistic updates so high-cardinality searches stay reproducible across devices.
- Designed secure **REST** endpoints with **Spring Security** (**JWT**, **CSRF**) and centralized exception handling; enforced bean validation, audit logging, and token rotation to harden auth, dynamic filtering, and password-reset workflows end-to-end.
- Tuned **MySQL** using composite/covering indexes, read replicas, and slow-query profiling; introduced **Redis** read-through cache with key-scoped invalidation, cutting p95 **API** latency by ~25% and stabilizing throughput in end-to-end canary runs.

TECHNICAL SKILLS & CERTIFICATES

Languages: Java, Go, Python, JavaScript/TypeScript, SQL, C/C++

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, EKS), GCP, **Docker**, **Kubernetes**, **Terraform**, **CI/CD**, **Linux/Unix**, **OpenTelemetry**, **Prometheus/Grafana**

Databases & Middleware: **PostgreSQL** (**pgvector**), **MySQL**, **Redis**, **Elasticsearch**, **MongoDB**, **Kafka**, **SQS**, **REST**, **gRPC**

Frameworks & ML: **Spring Boot**, **Node.js/Express**, **React/React Native**, **Flask**, **TensorFlow**, **scikit-learn**, **Pandas**

Certificates & Practices: IBM AI Engineering / ML, Google Data Analytics, **SOLID**, Automated Testing (**JUnit**, **Jest**, **PyTest**)